Marmara Üniversitesi
Electrical and Electronics Engineering Department
Fall 2023
Optoelectronics - Homework (1) – Due November 10, 2023

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1) Light reflection off single interfaces

Refer to the air-glass interface case discussed in class. Considering both TE and TM polarizations, produce the reflection curves ($|\Gamma|$ vs. θ_i) for the following cases. Use MATLAB. Discuss your results.

a) Green light impinging on ocean surface.
b) Components of white light incident on green leaves. Do your results support the fact that green leaves look "green"?

2) Light Wave Propagation in Homogeneous Media

Study the propagation of red light (wavelength=680 nm) inside Si material which has a complex refractive index. Specifically, find the complex dielectric constant, attenuation coefficient, wavenumber and complex impedance at the given wavelength. What percentage of light power will be absorbed within 100 μm of propagation inside this material?

Upload your solution file (pdf only) on the link below.

https://www.dropbox.com/request/KCffcfbT9p52CFMeRRb9